[RTU-520, 540, 560]

[ABB]

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1. INTRODUCTION.

1.1 GENERAL.

RTU it's a device has integrate with Data as Analog Inputs, Binary Inputs, Binary Output, Digital Input, Digital Output from any type of IED with the provision of Standard Protocol.

Device Name	RTU
Model Number	520, 540, 560
Product Make	ABB India Ltd
Configuration Tool	RTU till 500 Series

In What Ever the Types of RTU its 520, 540, 560 We Use Only the RTUtill Software, Software version Depends up on the firmware.

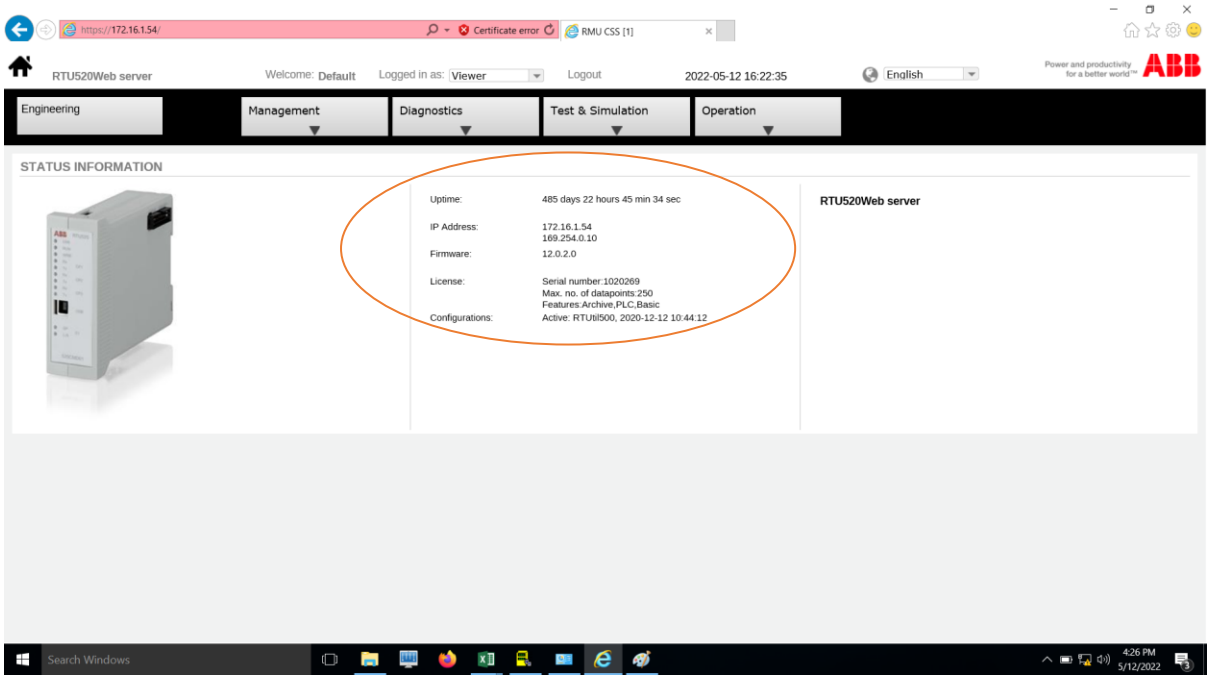


FIG 1.1

Data integration (either Hard or Soft) overview for the RTU as **Server/Client OR Master/Slave**

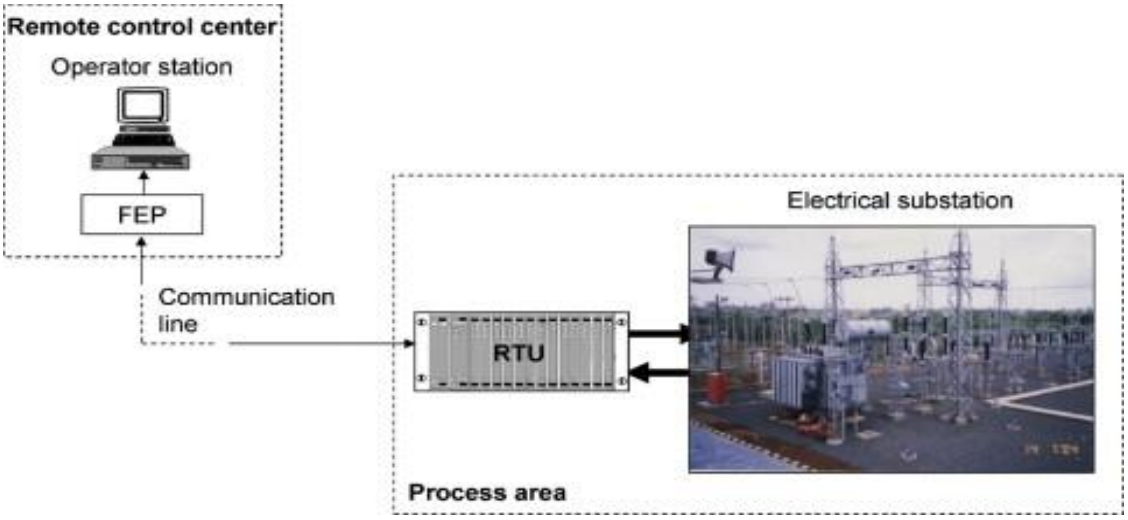
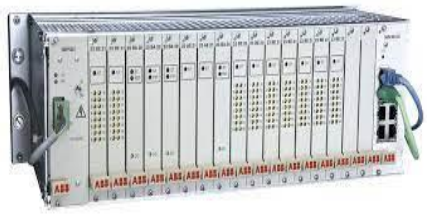


FIG 1.2

1.2 DEVICE INFORMATION.

In ABB three models of RTU device has been differentiate with their physical appearance

RTU520	RTU540	RTU560
		
1. Serial Bus Arrangement 2. Only 8 Cards Are Arrange in Single Rack except communication cards and power supply cards 3. Only Use Basic Licence As 250 Data Points & Upgrade 750 Datapoints	1. Serial Bus Arrangement 2. Only 8 Cards Are Arrange in Single Rack 3. Communication card has individual and IO cards are separate its has been connected with Bus connector. 4. Only Use Basic Licence As 250 Data Points & Upgrade 750 Datapoints	1. Arrangement of IO cards are fitted in the RTU 2. Licence basic is 5000 data points 3. Communication cards has installed as per the site requirement (max 2 Active /standby) 4. Body earthing has must

*****Whenever we start the configuration, we need to check the IO cards assembly. Configuration for hard signal as per the IO cards arrangement. *

2. RTU till CONFIGURATION

We have to use the RTU till software for the configuration of ABB RTU

Version check with firmware as same as RTU till version.

Ex:

firmware: 12.2.1.1

RTU till: 12.2.1.. to 12.2.0…

*Note: Licence and Tool version is must in same series **Refer 1.1***

In this RTU till we use three type of interface use to configure the project file.

2.1 Network tree.
2.2 Hardware tree.
2.3 Signal tree.

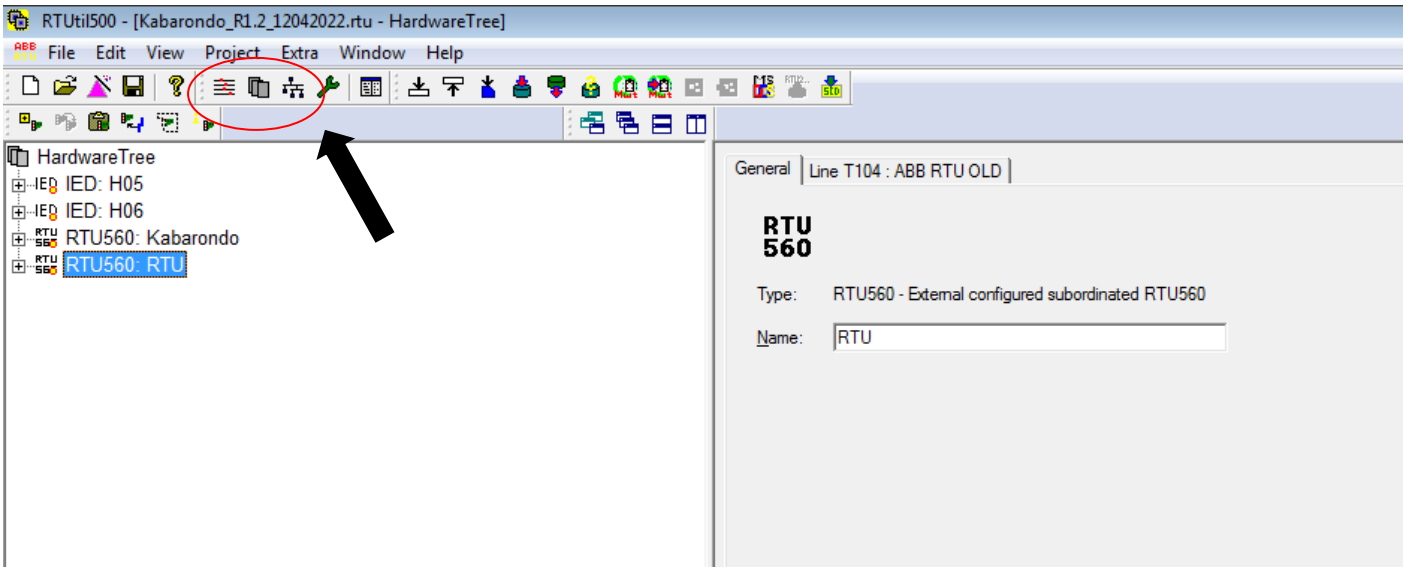


FIG 2.1

Note:

1. **Every change in the Project file must save the file**
2. **SAVE format is ".rtu"**
3. **Project generating file is ".rcd"**

2.1 NETWORK TREE

1.To assign the protocol either data Send OR Receive.

2. Integrate IED soft signals via standard protocols that's we need to add the device like MFM, Relay, Energy meter etc.

3. Data configuration using two Activity's for data send and receive (Host and Sub).

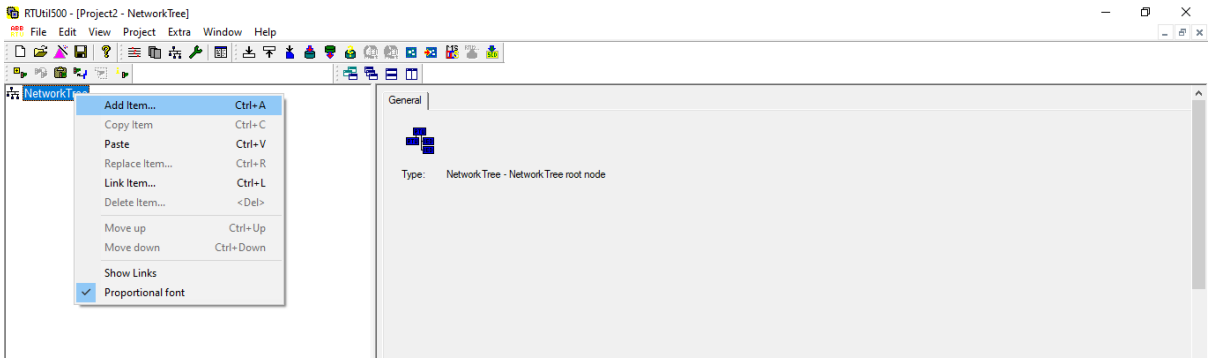
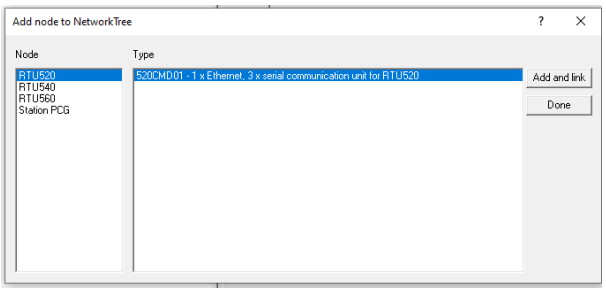


FIG 2.1.1

3. As per the device(RELAY, MFM, OR OTHERS) available in the site we need to add the IED for further configuration.
Add and link >> "RTU520 config add in the page"



Ex: RTU520,

FIG 2.1.2

4. Right Click the RTU 520 >> **Add >> Data need to send "add Host Activity"**
>>**Data need to receive "add Sub Activity"**

5. In this configuration Device or IEDs integrating with RTU via following protocol.
(* protocol has assigned as per the site requirement*)

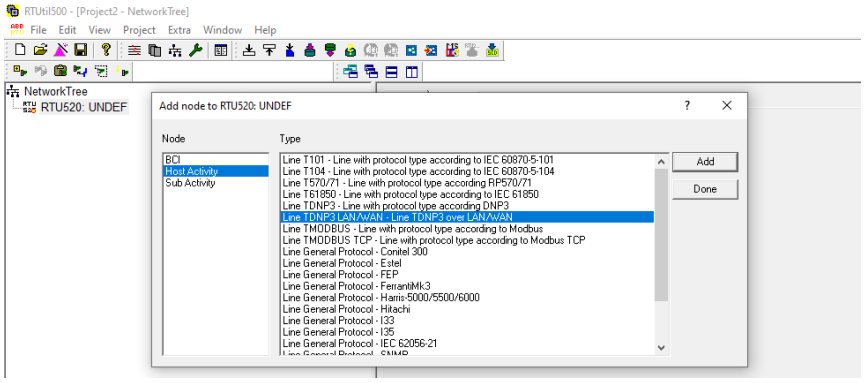


FIG 2.1.3

- For example only, Now I assign the configuration at **IEC61850** for data receive and **IEC60870-104** for data send.

6. Add 2 lines for data send and receive

Right click RTU520 >>ADD >>Host activity >> "**T-104 Master line**"
Right click 104 line >> **Control System** >> Add and Link.

Master line configuration has need to set the remote end IP Address (as per the site)

Right click RTU520 >>ADD >>Sub activity >> "**Line -T 61850**"
Right click **Line -T 61850** >> **IED**

In 61850 data we are received soft signal. So, IED (either relay or MFM) need to be added.

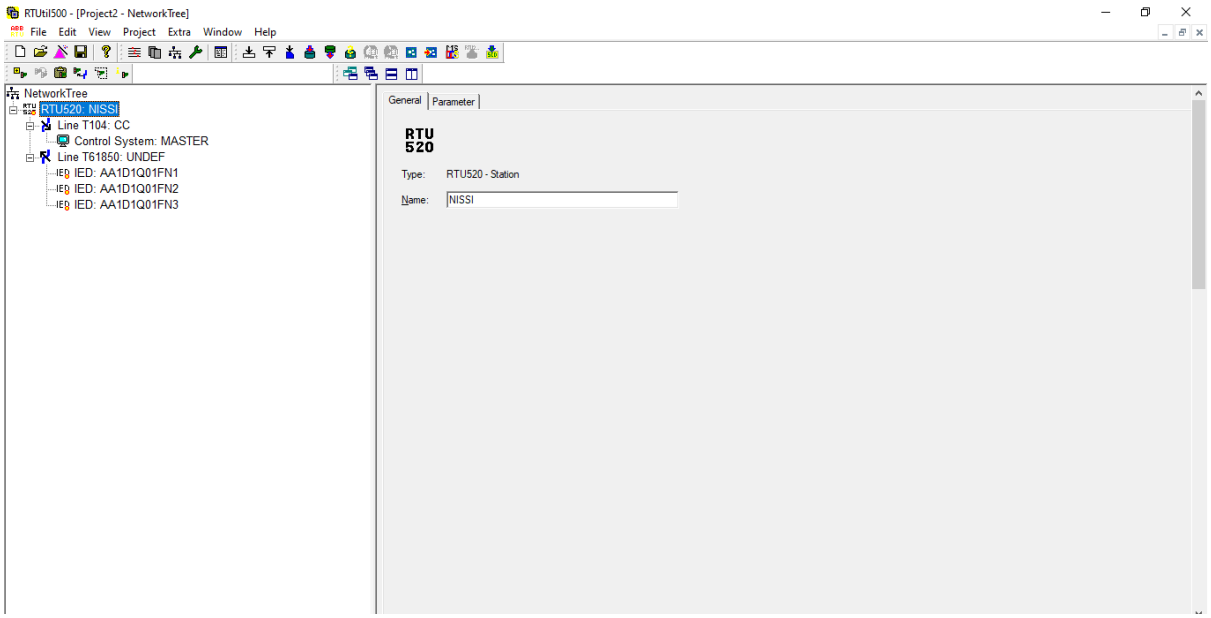


FIG 2.1.4

2.2 HARDWARE TREE

Wherever the IEDS are configure in the network tree it showing in hardware tree also

All the soft signals and hard signals are configured in the hardware tree only.
Right click **IED >> Add item.**

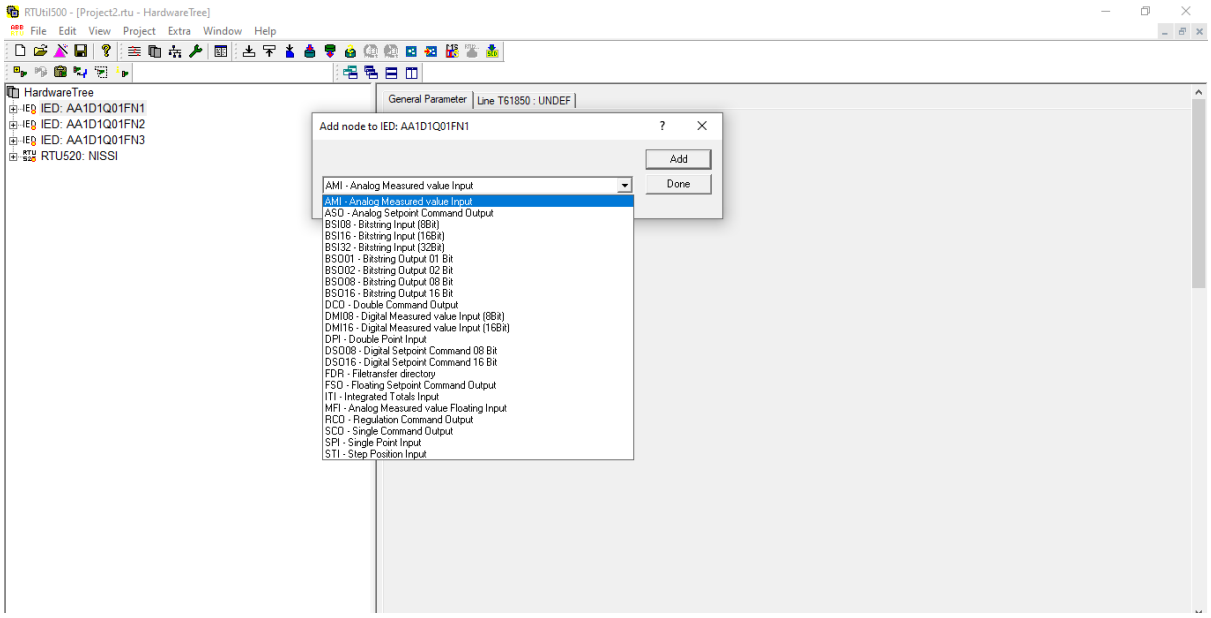


FIG 2.2.1

The following method to configure the soft signals in hardware tree as per the requirement.
After add the soft signals we need to assign **remote address**

////////////////////// *If its T-Modbus we use information object address.
If its T-61850 we use IEC node address (export in excel and assign).
If its T-104 or 101 we use ASDU and information object address. ////////////////////////////////////

2.2.1 RTU Hard signals assign,

All Hard signals are done by wiring by every card as per the inputs,

1. Analog input cards (AID01).
2. Binary input cards (BID01).
3. Binary output cards (BOD01).
4. Communication cards (CMD01).

////Refer 1.2 Device information//////

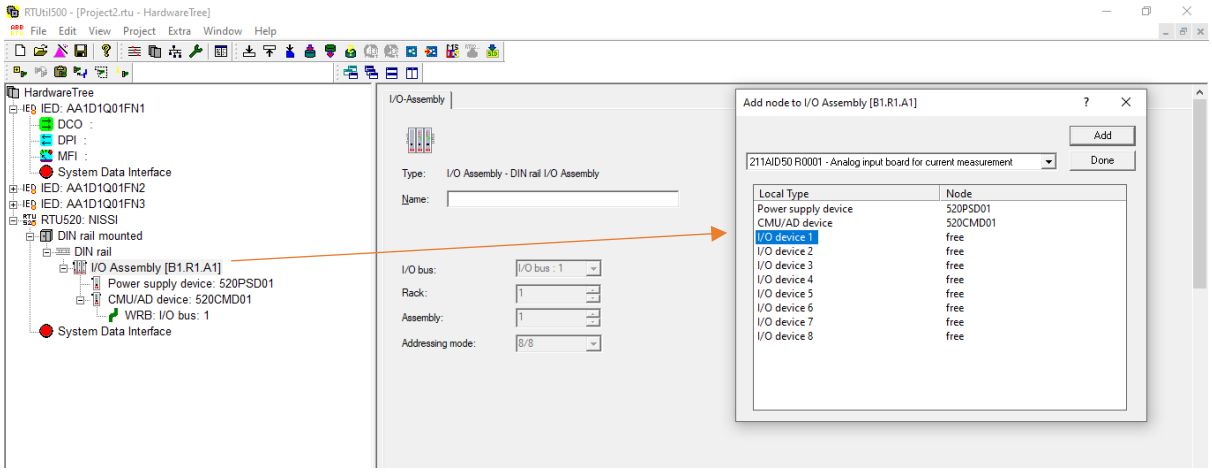


FIG 2.2.2

1. In RTU 520, IO Assemble only 8 cards has been attached except Power supply card and communication card.

2. Cards arrangement configuration as per the physical card's arrangement in RTU.

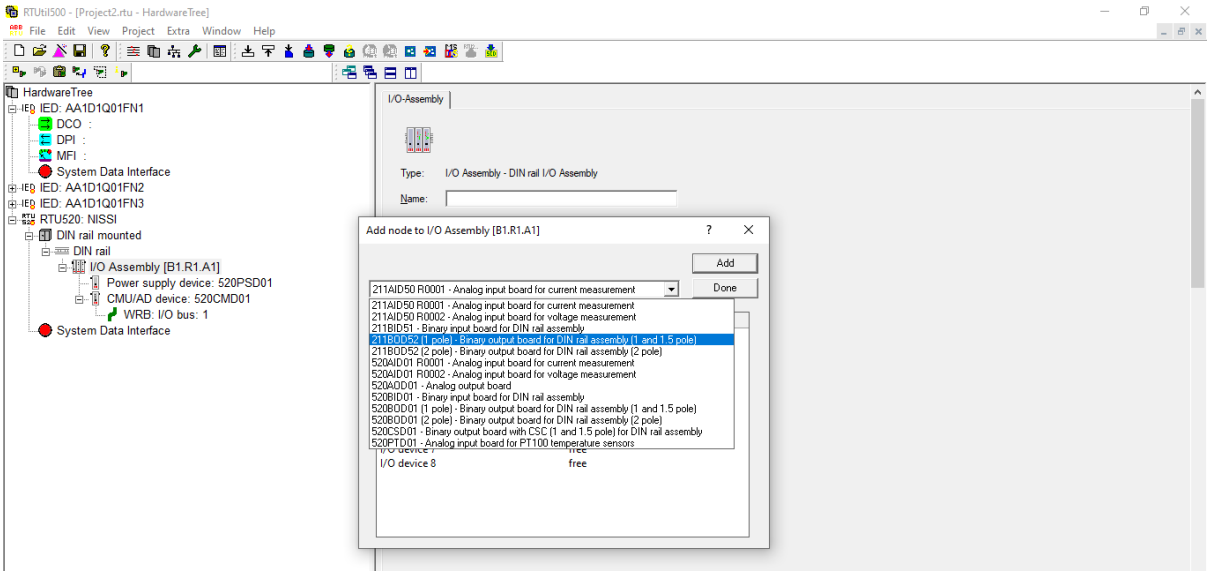


FIG 2.2.3

3. In this figure we add the Binary input & output card, Analog card
4. After complete the cards configuration in IO assembly, add the hard wiring signals.
   As per the signals type,
   DPI – Double point input.
   DCO – Double command output.
   SPI – Single point Input.
   SPO – Single Point Output.

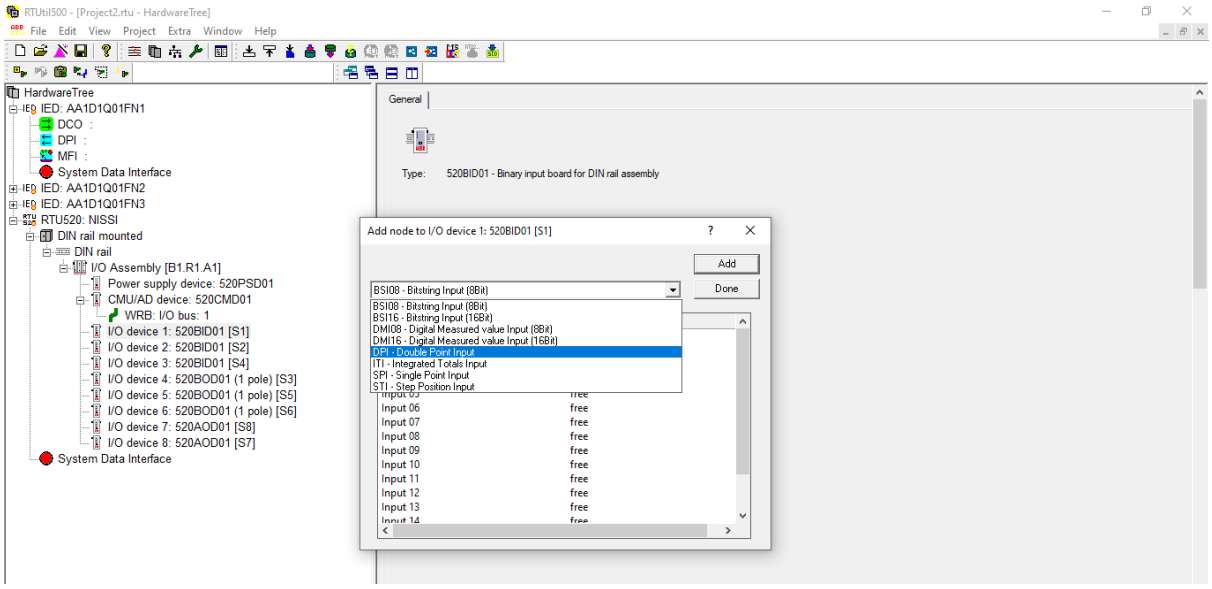


FIG 2.2.4

5. If its input cards add signals as per the scheme drawing. Depends up on the signal type we configure SPI or DPI
6. If its output card add signal as per the scheme drawing. Depends up on the signal type we configure SCO or DCO
7. If its Analog signal we should configure AMI or MFI signal as per the scheme
   Note: Input for the card is 4-20mA

2.2.2 Communication card

1. **Serials ports (CP1, CP2, CP3)**
   1. It's used to communicate the serial communication network devices.
   2. Its use RS232, RS485
   3. In the serial communication need to set the parameter settings.

***Parameters: Device ID, Baud Rate, Parity, Stop bit, Word length**.

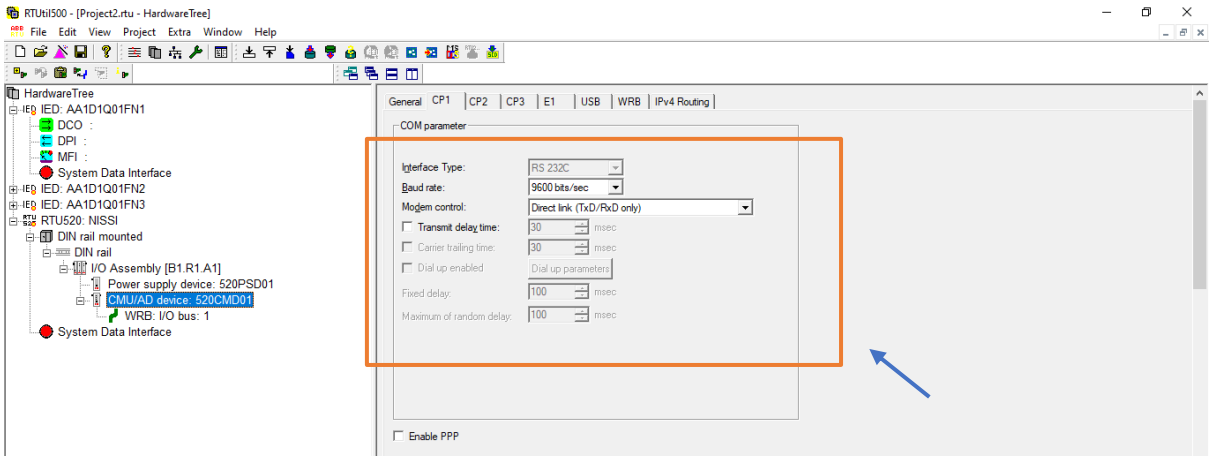


FIG 2.2.5

- RTU and Integrating Device is in same parameter setting or communication is not established.

2. Lan ports (E1, E2)

- To link the communication Protocol for Data, Send & Receive.
- In 520 only one communication, but 540 and 560 has 2 ports basically.

Right click CMD01 cards >>Link Item >> "to select the communication protocols were add in the network tree" >>select >>link.

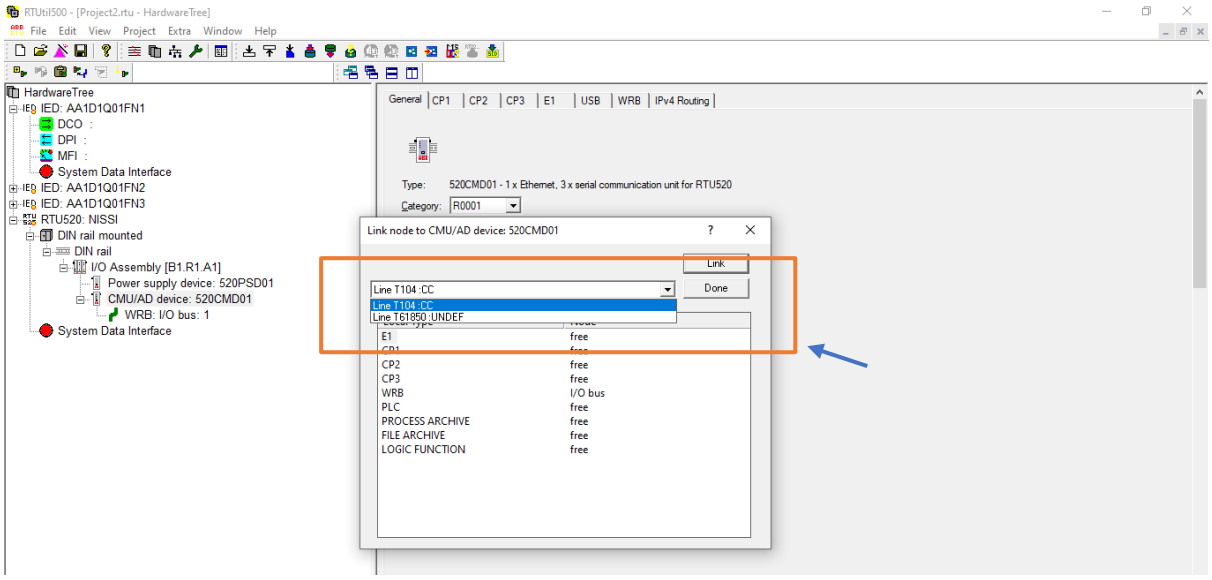


FIG 2.2.6

- In this communication port we should assign the IP address as per the site requirement.

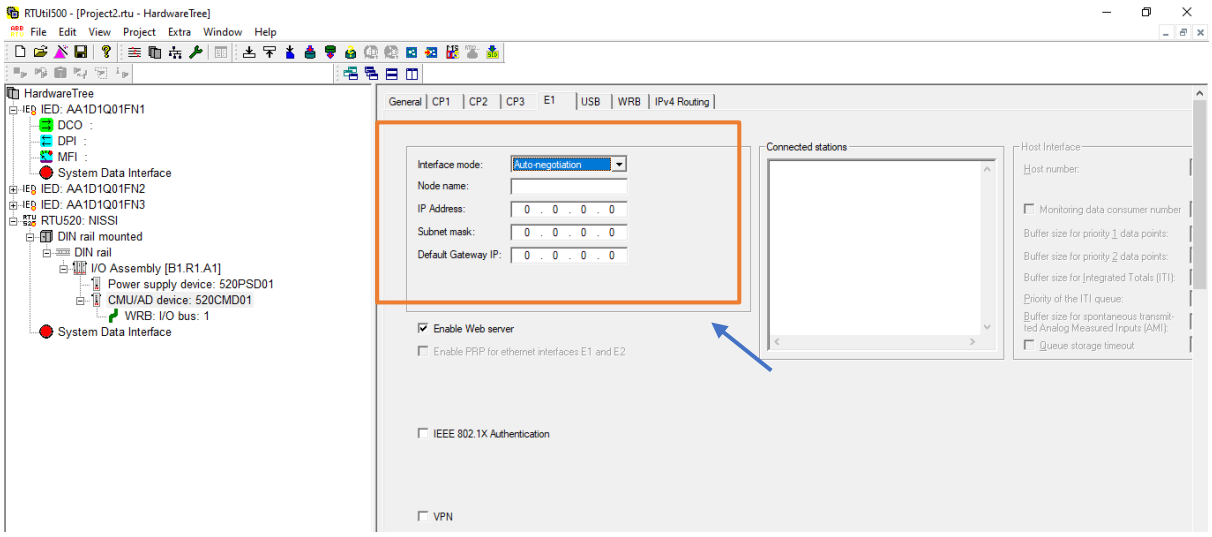


FIG 2.2.7

2.3 SIGANL TREE

Its used to arrange the all input, output signals analogue signals are arranged in sequence order

i.e. Site Name >> Substation Name >> Voltage level >> Bay Name >>" Link item(SPI, DPI, SCO, DCO, etc..) "

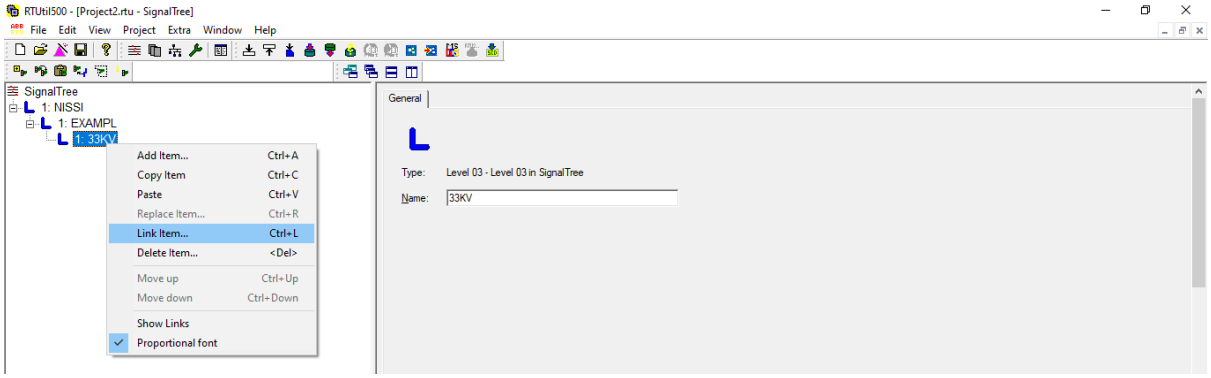


FIG 2.3.1

//////////////////////////////////////After the arrangement of Signal Tree//

3. PROJECT FILE GENERATION

To generate the RTU project file

1. Check consistency whether file have any error or Not
2. Click “PROJECT”

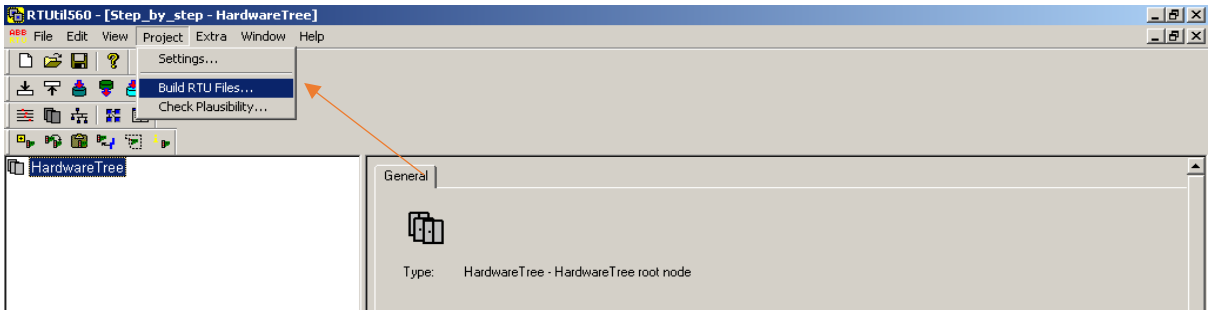


FIG 3.1

3. >>Built RTU file

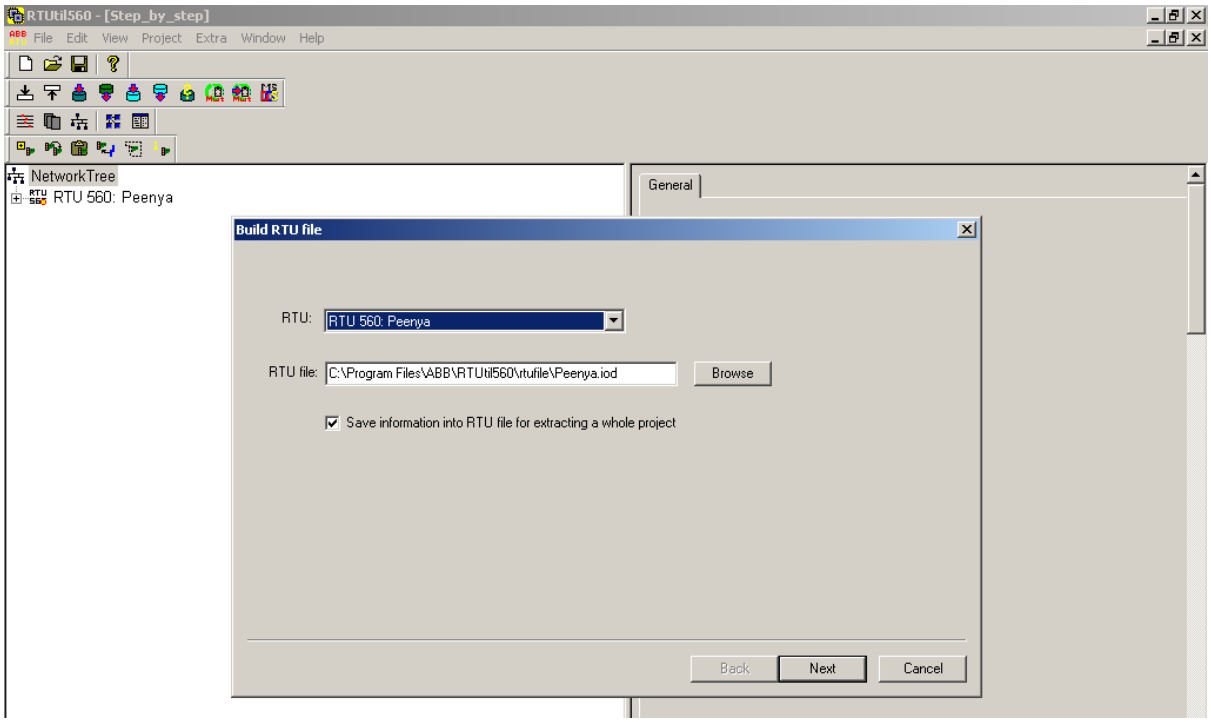


FIG 3.2

4. Click BROWSE and select the path location.
5. Save the Project.

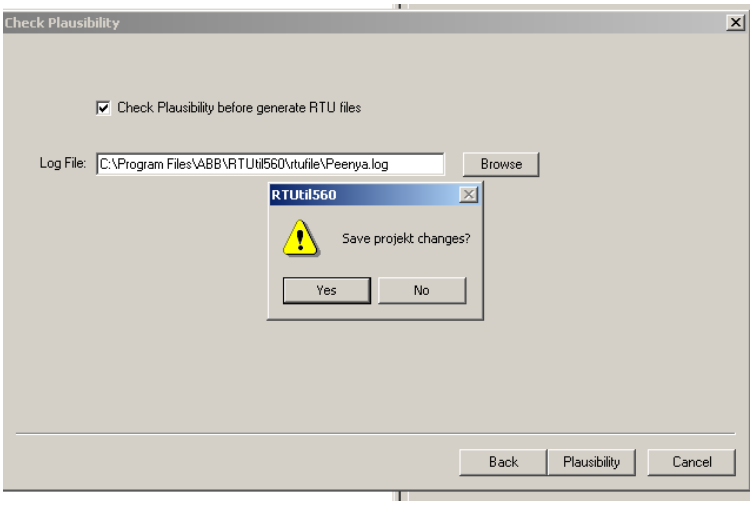


FIG 3.3

6. Generate the file “without any error”.

“Note: Save file format is .rtu / Project Generate file format is .rcd.”

4. SIMULATION AND TESTING

NOTE: USE FIREFOX FOR BETTER EXPERIENCE

Step 1: Open Firefox and type the RTU IP address and hit enter

Ex: https://192.168.0.1

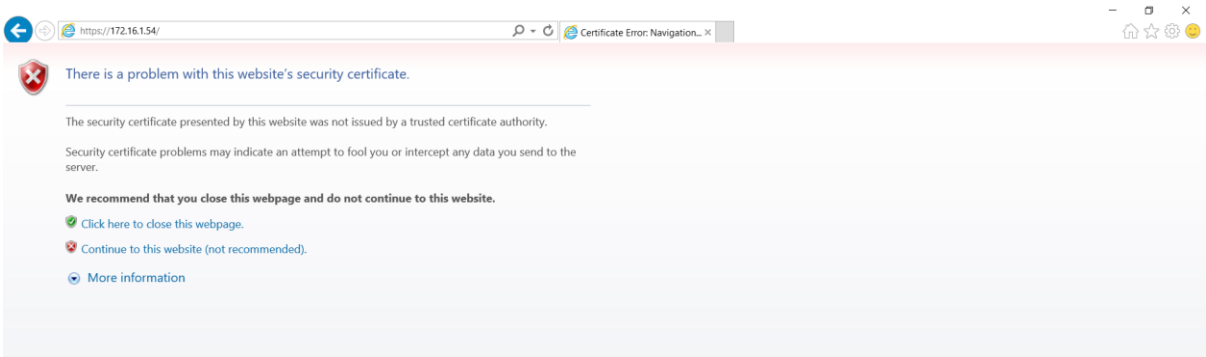


FIG 4.1

Step 2: click **"continue the website"** refer FIG 3.1

Step 3: Enter the Username / Password

User Name: Default

Password: Default

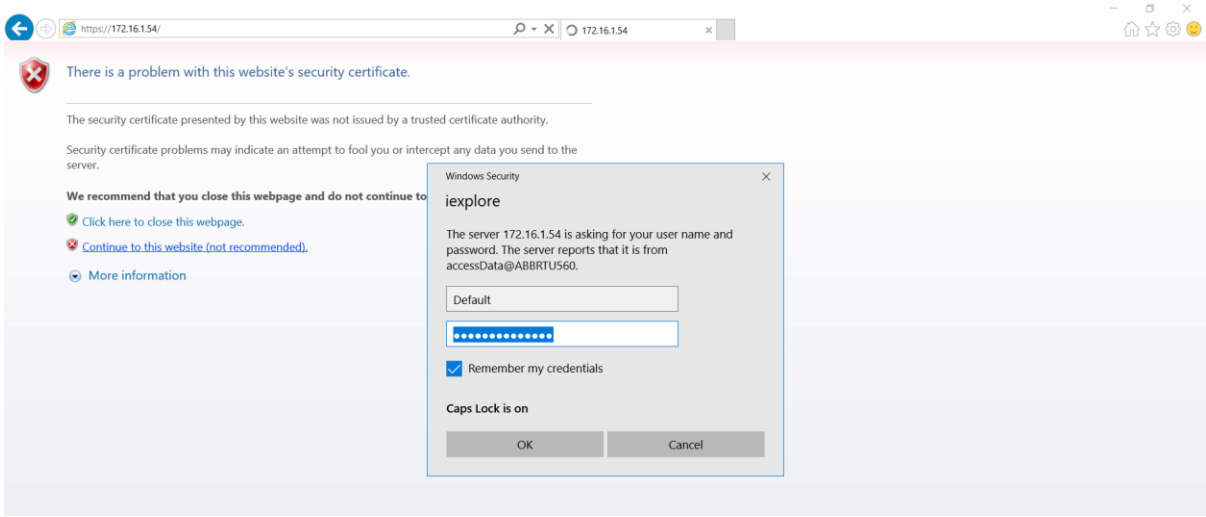


FIG 4.2

Username/Password: Default/Default; Admin/Admin; Root/Root

Step 4: Project file upload / Download

1. Change the user from VIEWER to ENGINEER.

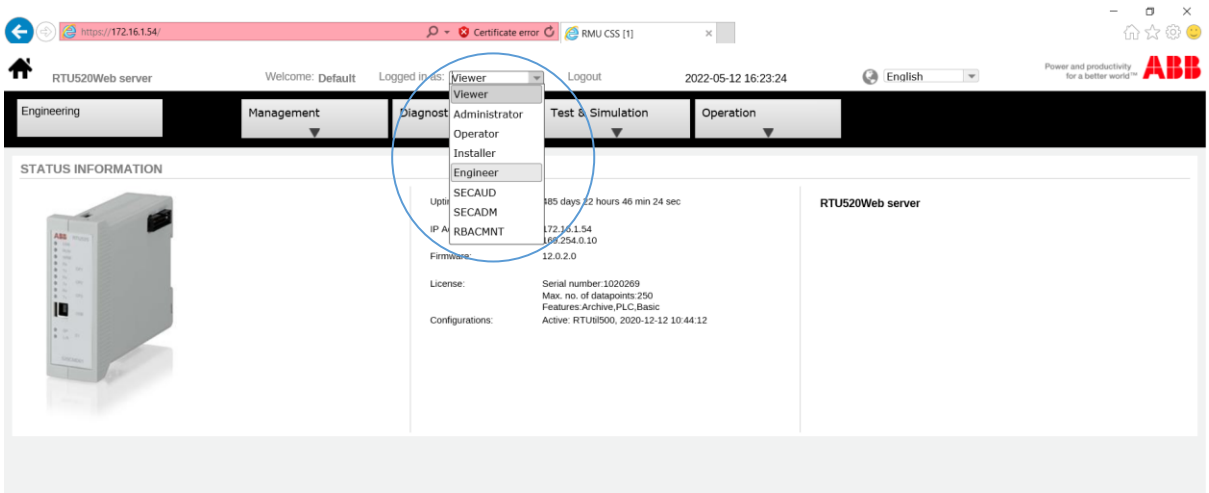


FIG 4.3

2. Click Management >> Configuration Management

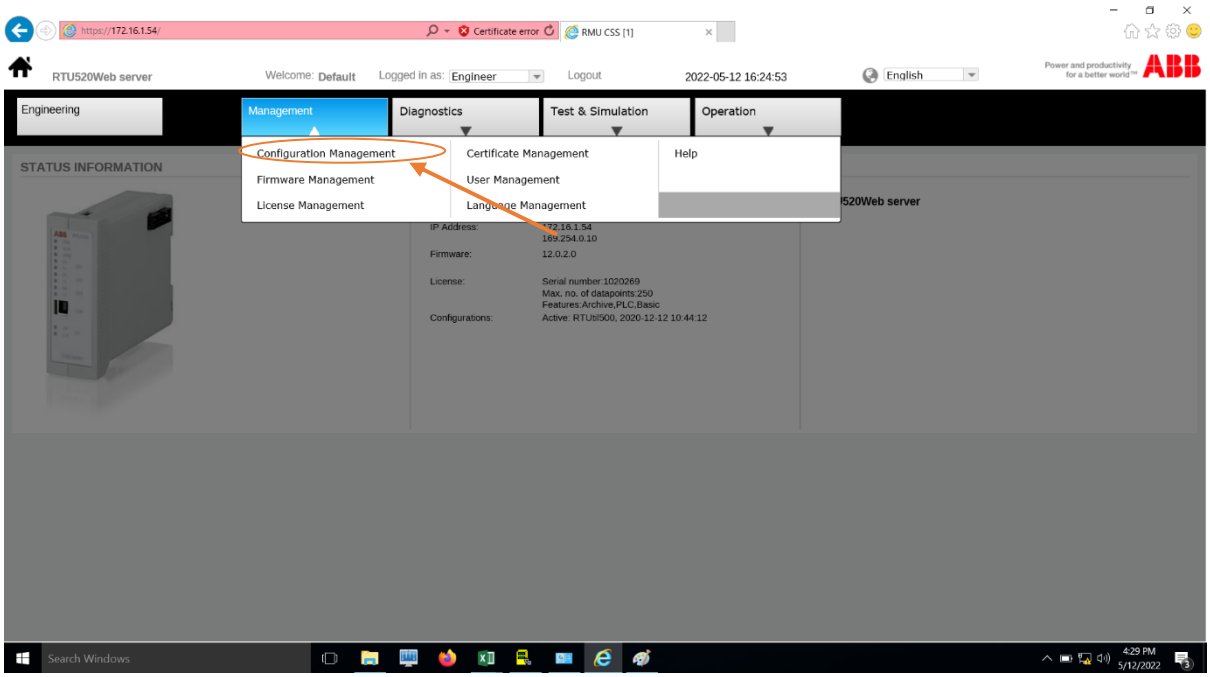


FIG 4.4

3. Download Existing Backup File / Load a backup file from RTU,

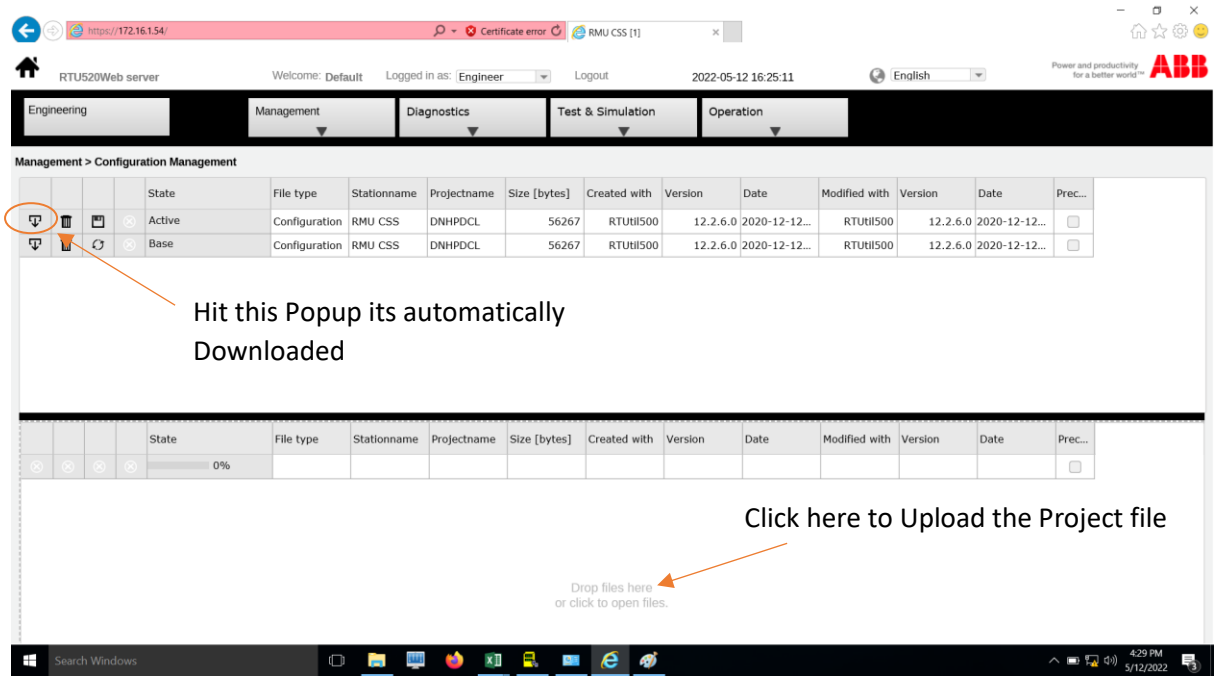


FIG 4.5

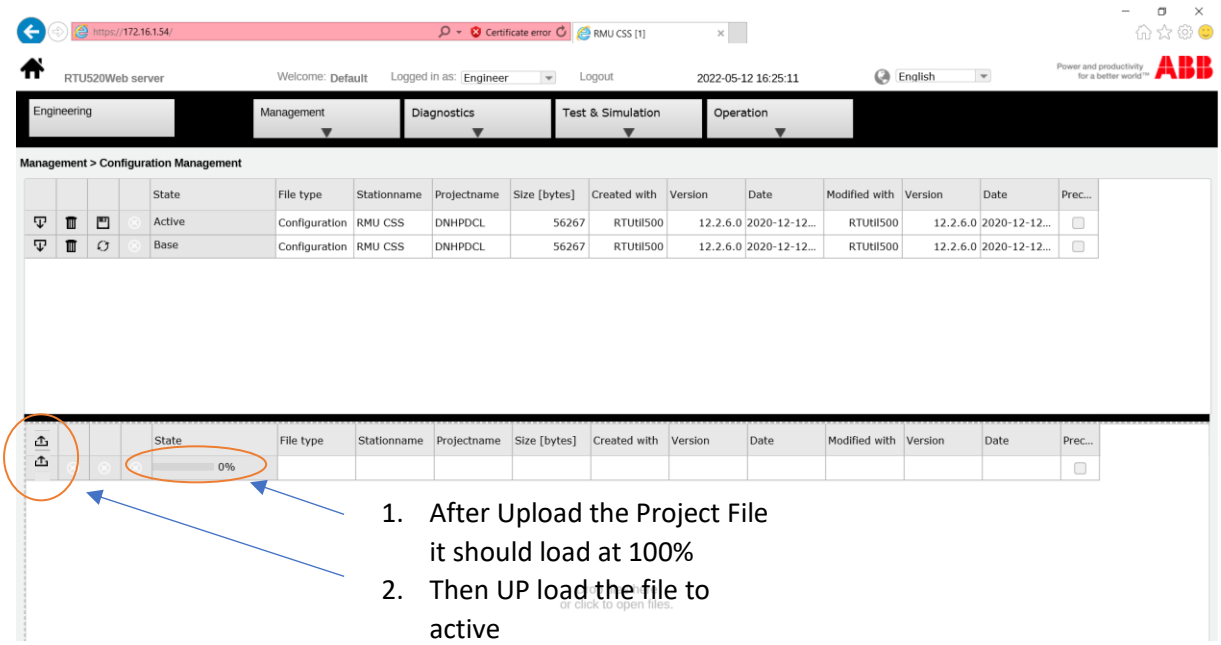


FIG 4.6

4. After the completion of TWO steps RTU has been Restated or Restart the RTU once.

STEP 5: To check the Configured indication in RTU backup.

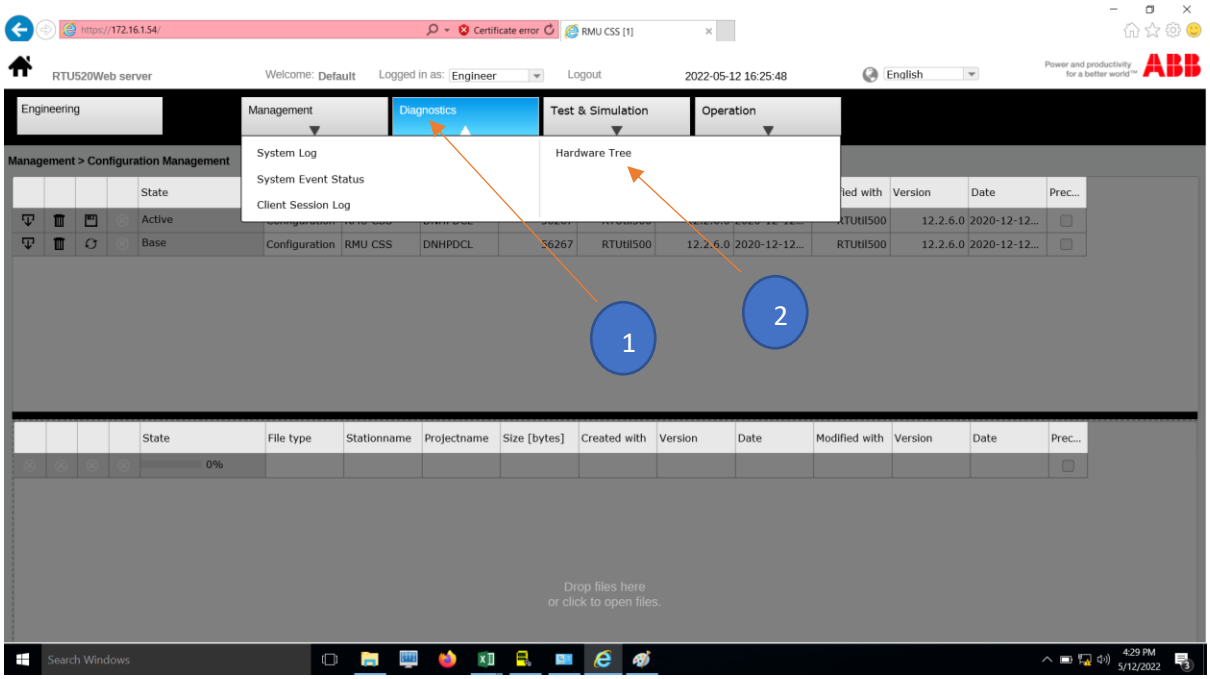


FIG 4.7

1. Follow the above points 1 and 2.

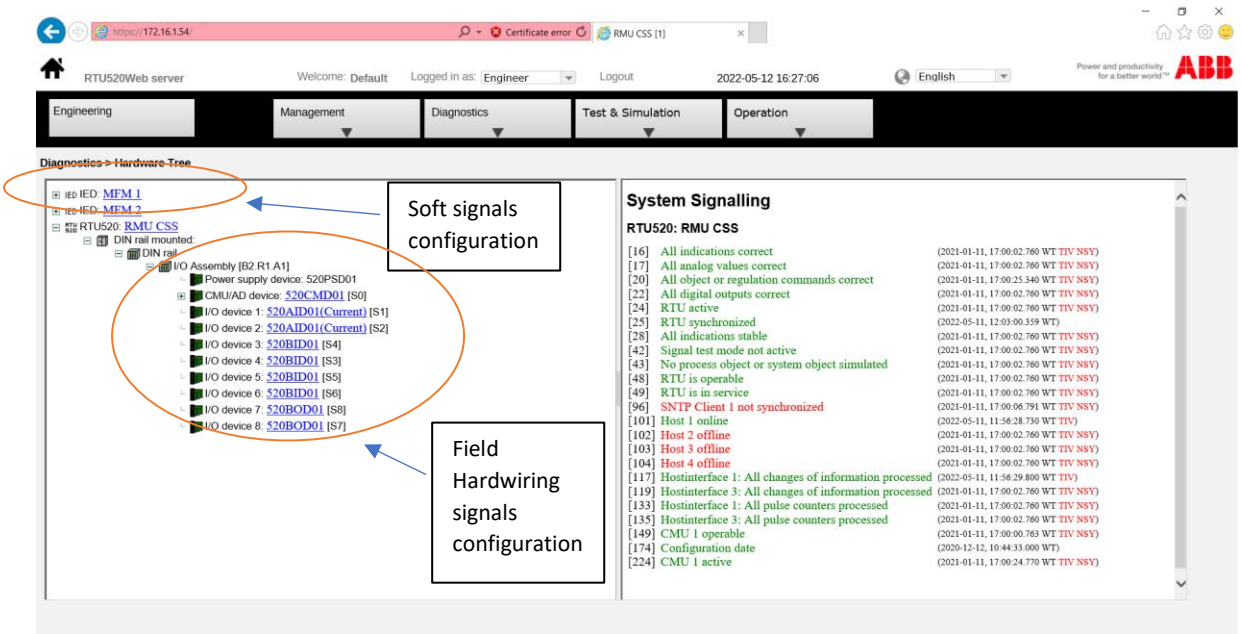


FIG 4.8

2. The above display page has Show All Binary inputs were its reporting or not.

3. Simulation or Command checking (Binary output)
4. Select **Testing and Simulation** tab in present display, then select the **Logging and debugging.**

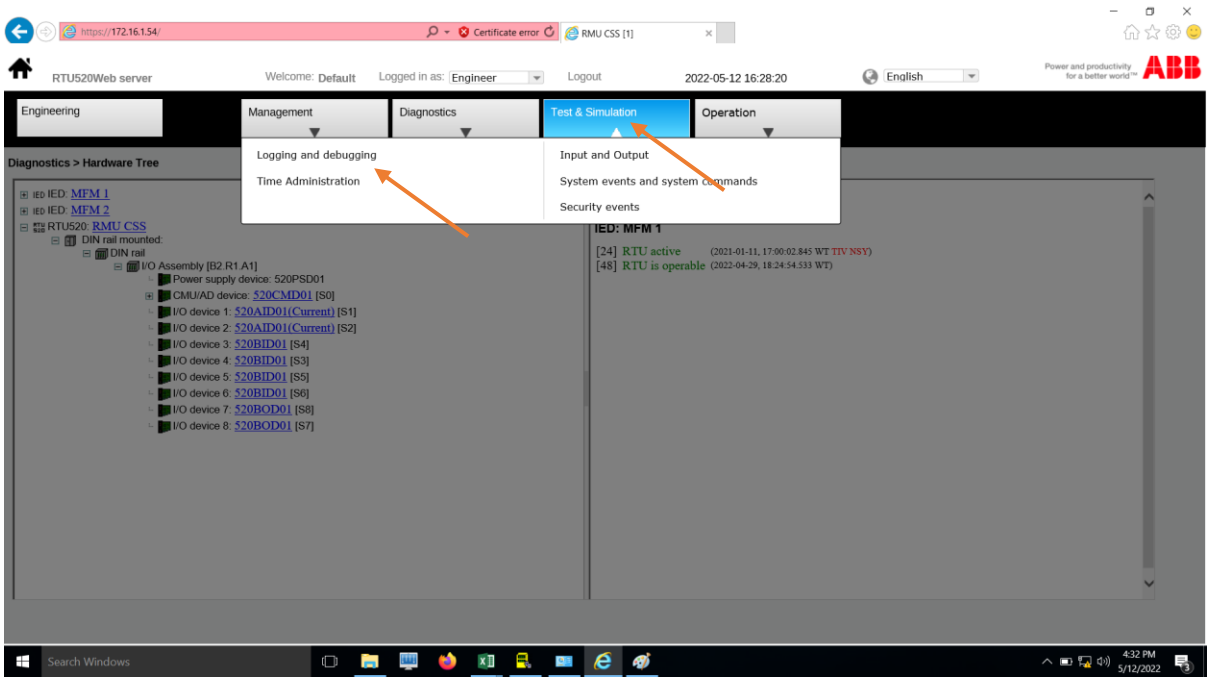


FIG 4.9

5. To enable the simulation in test mode.

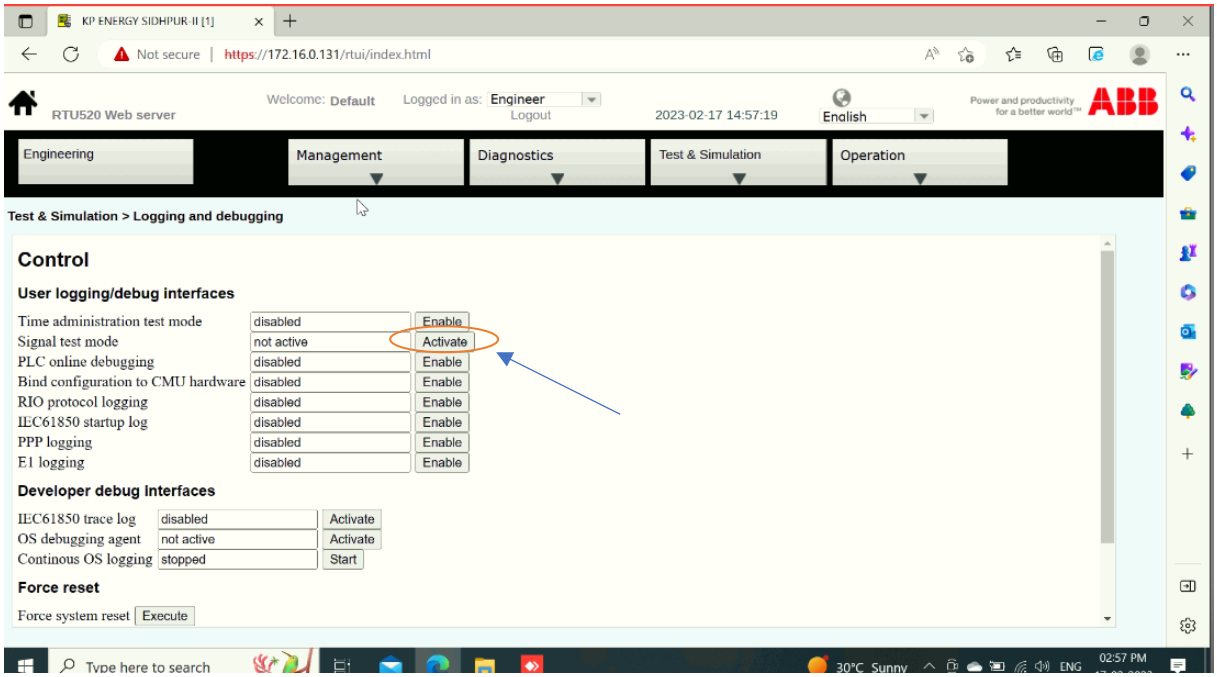


FIG 4.10

6. Again Select **Testing and Simulation** tab in present display, then select the **INPUT AND OUTPUT.**

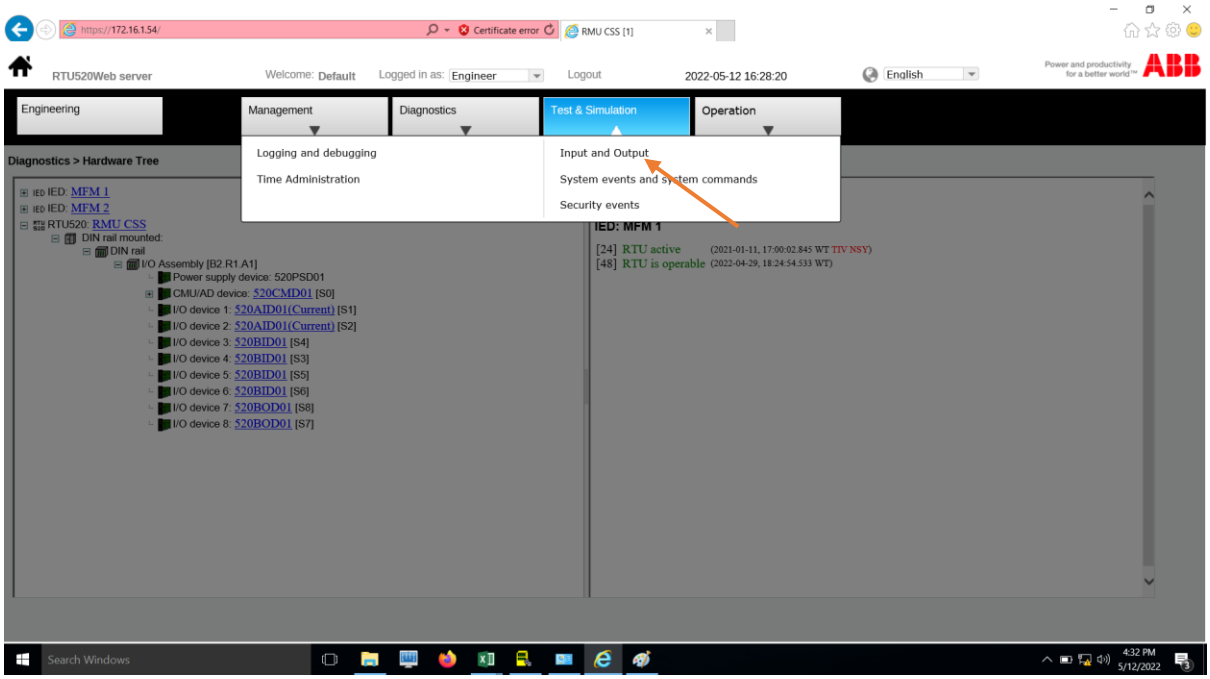


FIG 4.11

7. **To check the all signals here, set filter for command either SCO or DCO**

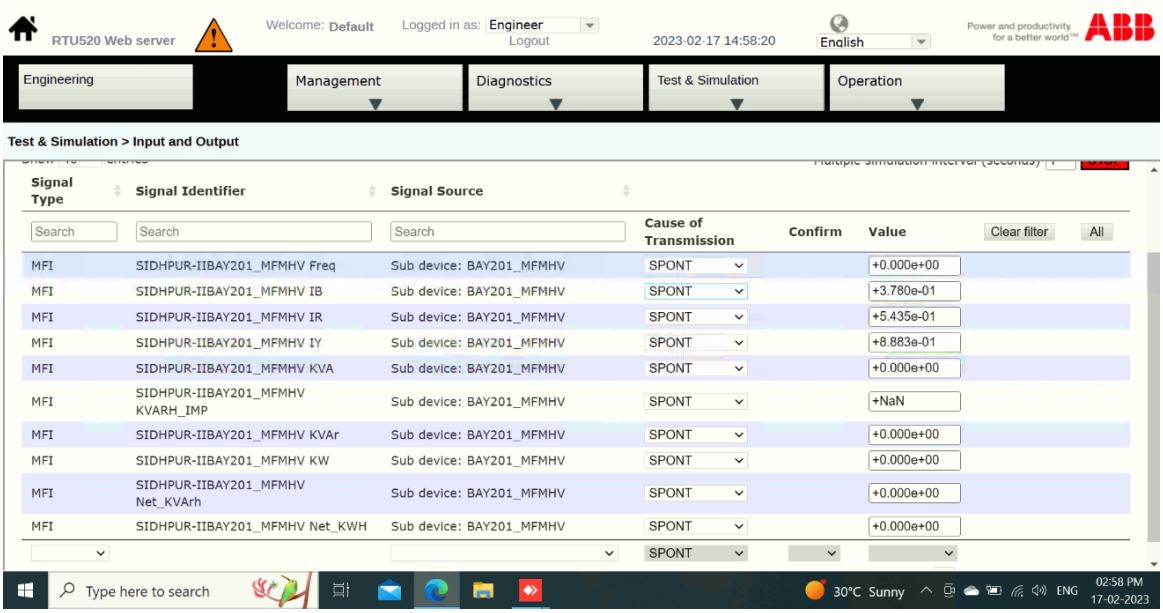


FIG 4.12

8. Refer the above pic Select **Case of Transmission** Select the Signals TYPE Whether its SCO or DCO.

//////////////After the completion of DO checking turn off the Simulation mode//////////////////